



ZHANG JIANNAN

PhD Candidate

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Munich, Germany

jiannan-zhang

STRENGTHS

Flight Physics & Control

Flight Modelling & Simulation

MBSE

Certification

Flight Testing

SOFTWARE SKILLS

MATLAB/Simulink

Control Design Toolbox

Math Symbolic Toolbox

Git

C/C++

LANGUAGES

English: Professional

Chinese: Native

German: Goethe B1

SUMMARY

Motivation: Productize eVTOLs.

Industry: 10+ years on eVTOL FCS, 1 TC, 1000+h flight testing.

Academy: 10+ publications, 300+ citations.

PROFESSIONAL EXPERIENCE

Research Associate | Institute of Flight System Dynamics, TUM

06.2016 – Now

Munich, Germany

- Lead of control allocation algorithm.
- Lead of control configuration optimization.
- Flight control system development.
- Flight dynamics modelling toolchain development.

Lead of Flight Control System | AutoFlight

08.2023 – 07.2024

Shanghai, China

- Lead of GNC development.
- Lead of system architecturing.
- Lead of FCS TC applications.

Senior Flight Control Engineer | AutoFlight

07.2021 – 08.2023

Shanghai, China

- Lead of guidance and control law development.
- Lead of flight dynamics modelling development.
- System architecturing.
- Certification tasks (requirement capture and validation).
- Flight testing.

Software Development | Phoenix-Wings GmbH

06.2020 – 07.2021

Ismaning, Germany

- Lead of linearization tool.
- Simulation toolchains development.
- Flight testing.
- Project management.

Flight Control Engineer | Autel Robotics Europe GmbH

06.2016 – 01.2020

Ismaning, Germany

- Flight control development for tilt-rotor UAV (Dragonfish).
- Attitude loop design and control allocation.
- Flight testing.

EDUCATION

Doctoral Candidate [Technical University of Munich \(TUM\)](#)

📅 11.2017 – Now (Expected Grad.: 2026) 📍 Munich, Germany

- **Research Direction:** eVTOL Control Authority and Allocation Optimization

M.Sc. Aerospace Engineering [Nanyang Technological University \(NTU\)](#)

📅 08.2013 – 02.2016 📍 Singapore

- **CGPA:** 4.92 (CAP: 1.06) **Major Subjects:** Flight Control, Light Weight Structures

B.Eng. Aircraft Design Engineering [Beijing Institute of Technology \(BIT\)](#)

📅 08.2008 – 07.2012 📍 Beijing, China

- **GPA:** 85/100 **Major Subjects:** Flight Mechanics, Aerodynamics

PROJECTS

V280 - 280kg eVTOL [TUM-FSD](#)

📅 02.2020 – Now DEV Industry Project

- **Role:** Lead of control allocation development
- **Main Contribution:** Develop the control allocation algorithm for the complete flight envelope (hover, transition and cruise flight). In addition to nominal operation, the algorithm can safely operate with a critical hover propeller failure given information from fault detection and monitoring scheme.
- **Development:** Strictly under code-compliant guidelines specified by DO-331 MBD design process (standard libraries, Model Advisor checks, Simulink Test Harnesses, and guaranteed WCET on target platform); automatic tests by continuous integration at push time & issue tracking with GitLab; requirements tracking in Polarion.

Extended F-16 Modelling and Control | [TUM-FSD](#)

📅 05.2015 – 07.2017 DEV Research Project

- **Role:** Lead of simulation model and baseline control design
- **Main Contribution:** Derive & build the simulation model and closed-loop framework. Verified the INDI concept and compared several different control allocation schemes.
- **Objective:** The highly redundant and nonlinear model serves as a conceptualization platform before a strategy is utilized to industry projects.
- **Development:** The closed-loop framework is setup in MATLAB/Simulink with Flight-Gear visualization. Implementation details are found in Publications [1][3].

Dragonfish - 5kg Tilt-rotor Transition UAV | [Autel Robotics Europe GmbH](#)

📅 06.2016 – 01.2020 DEV Industry Project

- **Role:** Lead of control allocation development
- **Main Contribution:** Developed the control allocation scheme for the entire envelope. Deeply involved in the autopilot design of attitude and velocity control, debugging and tuning the closed-loop behavior. The controller provides a smooth transition and unified behavior between hover and wingborne flight.
- **Development:** Deploys MATLAB/Simulink and code generation tool chain with some hundred hours of on-site flight test and tuning. By spring 2020, the complete autopilot software is matured for product pre-release.
- In addition, extensively contribute to administrative communication processes in the cooperation between the university team and the Chinese headquarter.

Manta Ray - 30kg eVTOL Delivery Drone | [Phoenix-Wings GmbH](#)

📅 06.2020 – Now DEV Industry Project

- **Role:** Programmer of Linearization Tool in C++
- **Main Contribution:** Design and implement the linearization tool in C++, as well as the MATLAB/Simulink interface for model verification and further design tasks.

- **Development:** Use generated C code of the Simulink model as interface. Code in Linux OS with open source Eclipse IDE and CMake projects. Use Ipopt and lapack library as trim solver.
- Additionally participate in sensor calibration processes and extensive software and on-site flight testing for the product tool chain; support in administrative processes with Chinese customers.

FIRST AUTHOR PUBLICATIONS

1. *Attainable Moment Set Optimization to Support Configuration Design: A Required Moment Set Based Approach*
2. *Saturation Protection with Pseudo Control Hedging: A Control Allocation Perspective*
3. *Control Allocation Framework with SVD-based Protection for a Tilt-rotor VTOL Transition Air Vehicle*
4. *Modeling and Incremental Nonlinear Dynamic Inversion Control for a Highly Redundant Flight System*
5. *Control Allocation Framework for a Tilt-rotor Vertical Take-off and Landing Transition Aircraft*